COMP 9200
Applied Research Methods

Qualitative methods

Overview

- Introduction
- Kinds of qualitative studies:
  - Case study
  - Ethnography
  - Phenomenological
  - Grounded Theory
  - Narrative Inquiry
  - Content Analysis
  - Discourse Analysis
- Data collection
- Interviews
- Data organization and analysis
- Biases

© Borna Noureddin

Introduction

- A number of qualitative methodologies
- All have two things in common:
  - Focus on phenomena that have occurred or are occurring
  - Involve capturing and studying the complexity of those phenomena
- Researchers recognize multifaceted form of issue they are studying
- Often the starting point of any inquiry: helps define what needs to be studied

© Borna Noureddin

Introduction

- The researcher is the instrument
- Methodology: often iterative and recursive process
- Formulate only general research problems and ask only general questions about phenomena to study
- May not identify ahead of time exact methods to use
- Methodology may continue to evolve over course of study
- Requires considerable preparation/planning
- Studies do not allow researcher to identify cause-and-effect relationships

© Borna Noureddin

Introduction

- Approaches in qualitative studies (different from quantitative)
  - May formulate only general research problems and ask only general questions about the phenomenon of interest
  - Questions become more specific as the study proceeds
  - Methodology may evolve over the course of the investigation
- Skills used in qualitative research
  - Firm grasp of previous research related to the problem
  - Training in observation techniques, interview strategies, and other data-collection methods
  - Ability to find meaning in huge amounts of data

© Borna Noureddin

Introduction

- Potential advantages:
  - Exploration: initial insights into topic or phenomenon
  - Multifaceted description: complex nature of the area of focus
  - Verification: test claims, theories, etc. in real-world settings
  - Theory development: development of new concepts, theoretical perspectives
  - Problem identification: uncover key problems
  - Evaluation: judge effectiveness of particular policies, practices, and innovations

© Borna Noureddin

Introduction

Kinds of qualitative studies:

- Case Study
- Ethnography
- Phenomenological studies
- Grounded Theory Study
- Narrative inquiries
- Content Analysis
- Discourse analyses

© Borna Noureddin

Case Study

- Researcher looks at one or a few individuals, programs, or events
  - In-depth study over an extended time period
  - Collection of several different kinds of data
  - May include observations, interviews, technology-based communication, documents, records, audiovisual materials
  - Includes portrayal of context surrounding the case
- Investigation may take into account broader physical, historical, cultural, social, and economic contexts
  - Consideration for their effects on the individual(s), program, or event(s) being studied

© Borna Noureddin

Case Study

Data Analysis:

- organization of details about case
- categorization of data
- interpretation of single instances
- identification of patterns
- synthesis and generalizations

© Borna Noureddin

Case Study

Research Report:

- rationale for studying case
- detailed description of related facts
- description of data collected
- discussion of patterns found
- connection to larger scheme of things

© Borna Noureddin

Ethnography

- Researcher examines entire group that shares a common culture
  - Usually for a lengthy period of time
  - Within the group's natural environment (site-based fieldwork)
  - Identifies primary informants
- Focuses on group members' everyday behavior
  - Identifies cultural norms, beliefs, practices, and social structures
- Researcher's role: passive observer or willing participant
- Data: extensive field notes, ongoing journals with reflections
- Autoethnography: research on one's own perspectives, beliefs, experiences, and practices

© Borna Noureddin

Ethnography

Data Analysis:

- collection and analysis occur somewhat simultaneously
- description
- analysis
- interpretation
- strives for rigorous subjectivity

© Borna Noureddin

Ethnography

Research Report:

- often written in personal, literary style
- introduction providing rationale for study
- description setting and methods
- analysis of group studies
- conclusion

© Borna Noureddin

Phenomenological Study

- Goal: understand people's perceptions and perspectives about a particular situation
  - Main question: "What is it like to experience such-and-such?"
- Depend almost entirely on lengthy interviews
  - Work with small, carefully selected sample of participants
  - May use three-interview series
- Suspend preconceived notions that unduly influence what they "hear" participants say in interviews

© Borna Noureddin

Grounded Theory Study

- Researcher begins without any theoretical framework in mind
  - Purpose: to construct a theory about a little-understood phenomenon
  - uses prescribed set of procedures for analyzing data and constructing theoretical model
  - theory "grounded" in data
- Typical focus: the nature of people's actions/interactions
  - Includes participants' perspectives about those actions/interactions
- Study begins with data analysis
  - Categorization of incoming data; speculation about possible concepts and themes
- Method: data collection is field-based, flexible, and likely to change over course of study
  - interviews typically play major role
  - constant comparative method is used: data analysis drives later data collection

© Borna Noureddin

Grounded Theory Study

Data Analysis:

- Includes:
  - open coding
  - axial coding
  - selective coding
  - development of a theory

- No matter what form theory takes, based entirely on data collected

© Borna Noureddin

Grounded Theory Study

The Research Report:

- writing is objective and impersonal and includes:
- description of research question
- review of related literature
- description of methodology and data analysis
- presentation of theory
- discussion of implications

© Borna Noureddin

Narrative Inquiries

- Method for studying complex, multifaceted phenomena
  - Relies on people's recollections and stories about their experiences related to these phenomena
    - Past and present interactions with others within the context
    - Personal rather than shared experience (unlike phenomenology)
- Three-dimensional narrative space
  - Concepts of interaction, continuity, and situation
- Results: researcher's rich narrative
  - Provides better understanding of setting, context, time period, and/or sequence of the phenomenon

© Borna Noureddin

Content Analysis

- Detailed, systematic examination of contents of a body of material
  - Typically, a form of verbal or nonverbal communication
  - Purpose: identify general patterns, themes, biases
- Begins with a main research question
  - Identifies sample materials, defines specific characteristics or qualities to examine
  - Scrutinizes each piece and imposes category labels
  - Tabulates frequencies of occurrences of each category (quantitative element)

© Borna Noureddin

Content Analysis

Method: systematic and includes

- identification of material to be studied
- definition of characteristics to be studied
- breakdown of complex items into smaller segments
- scrutiny of material for identified characteristics under study

© Borna Noureddin

Content Analysis

Data Analysis:

- involves tabulation of frequency of each characteristic found in material studied
  - tabulations and statistical analyses used to interpret data

Research Report:

- description of material studied
- precise definitions of material characteristics
- coding or rating procedures
- tabulations for each characteristic

© Borna Noureddin

Discourse Analysis

- Similar to content analysis
  - Examines contents of human communications
- Focus: language-based communications—discourse
  - Also examines other inherent messages
    - Body language, use of dialect; situational, personal, social, and cultural contexts of the discourse
- Constructivist perspective
  - Meanings presumed to be constructed by the listeners, readers, and researcher

© Borna Noureddin

Data Collection

- Multiple forms of data
  - Observations, interviews, objects, written documents, audiovisual materials, electronic entities, and more
- Memos: reflective, methodological, analytical
- Emergent design: often used in qualitative studies
  - Data collected early influence methods subsequently used and the kinds of data that are subsequently collected
  - Can be a disadvantage in constructing well-thought-out plan to present to an institutional review board (I R B)

© Borna Noureddin

Data Collection

- Data collected early in investigation may influence subsequent data
- Potential sources of data are unlimited
- Data collection takes a great deal of time
- Data collection methods should be consistent with ethical principles of research studies

© Borna Noureddin

Data Collection

- Seek credibility and transferability
- Strength of qualitative research: researcher is an instrument
  - Pros (discern underlying patterns) and cons (biases)
- Strategies to enhance trustworthiness and credibility of data
  - Reflexivity: regularly looking inward to identify personal biases
  - Triangulation: collecting multiple forms of data
  - Spending considerable time on site: immersing oneself in the field
  - Intentionally seeking exceptions and contradictory evidence
  - Seeking other people's feedback about data and analyses
  - Maintaining an audit trail

© Borna Noureddin

Data Collection

- Sampling:
  - choice of sample depends on research question
  - selection of data sources tends to be nonrandom
  - purposeful
- Observations:
  - researcher may be outsider/participant observer
  - intentionally unstructured and free-flowing
  - can be problematic and lack objectivity
- Interviews:
  - can yield great deal of information
  - tend to be unstructured and open-ended
  - occasional use of focus groups

© Borna Noureddin

Data Collection

- Purposive sampling in most qualitative studies
  - Choosing particular people or objects
- Grounded theory studies may use theoretical and discriminant
- Ethnographers use primary informants
- Case studies use extreme case sampling
- Some use convenience sampling or snowball sampling
- Size of sample: enough to provide information about variability that exists in the phenomenon under investigation

© Borna Noureddin

Data Collection - interviews

- Characteristics of qualitative interviews
  - Not as tightly prescribed/structured as in quantitative studies
  - Tend to be informal and friendly
  - May involve homogeneous or heterogeneous focus groups (interviewing several participants simultaneously)

© Borna Noureddin

Data Collection - interviews

- Identify general interview questions and possible follow-up subquestions in advance
- Consider how participants' cultural backgrounds might influence their responses
- Make sure your sample includes people who will give you the kinds of information you are seeking
- Find a suitable location
- Get written permission
- Establish and maintain rapport
- Focus on the concrete and actual rather than on the abstract or hypothetical

© Borna Noureddin

Data Collection - interviews

- Don’t put words in people’s mouths
- Record responses verbatim
- Keep your reactions to yourself
- Remember that you are not necessarily getting the facts
- When conducting a focus group, take group dynamics into account
- At the end of the interview, ask for any additional comments that participants might want to make

© Borna Noureddin

Data Collection - interviews

- Audio and video recordings of interviews
  - Options with tablets, smartphones, and camcorders downloaded to computer
- Voice recognition software for transcriptions (Hyper T R A N S C R I B E, for example)
- Internet-based for remote interviews
  - Zoom, Skype, and more video conferencing options
  - Chat rooms or bulletin boards for focus groups

© Borna Noureddin

Data organization and analysis

- Employ a data analysis spiral:
- Organize data
- Peruse data to get sense of what it contains as a whole
- Identify general categories or themes (and possible subthemes), then classify data accordingly
- Integrate and summarize the data

© Borna Noureddin

Biases

- Researcher bias
  - Personal experiences, perspectives, and worldviews
  - Influence which data are collected, impose inaccurate meanings, and may alienate participants
- Sampling bias
  - May not accurately capture a larger group from which the study should draw conclusions
- Response bias
  - Participants may provide inaccurate descriptions

© Borna Noureddin

Review

- Introduction
- Kinds of qualitative studies:
  - Case study
  - Ethnography
  - Phenomenological
  - Grounded Theory
  - Narrative Inquiry
  - Content Analysis
  - Discourse Analysis
- Data collection
- Interviews
- Data organization and analysis
- Biases

© Borna Noureddin